Gerard Lawler

gelawler | Gerard Lawler | gelawler | gelawler@protonmail.com | +1.508.577.6026

Summary

I am an early career physicist interested in applications of physics to industrial applications. I have extensive experience with coupled multiphysics simulation and experimental implementation including novel hands-on hardware development. My primary areas of experience lie in photoinjectors and electron linacs, radiofrequency devices using advanced material cathodes, for use in high brightness electron particle accelerators. As such, I have extensive experience in radiofrequency, pulsed power, and optical design. Notably, I have also studied surface electron emission behavior in extremely high electric fields and cryogenic temperatures with development of simulation and experiment as well as the use of plasma physics to understand high brightness electron beams.

Skills

Physics simulation	Ansys HFSS, CST, Lumerical, General Particle Tracer (GPT), Astra, Geant4
Programming	Python, C++, JAVA, Fortran
Other Software	SolidWorks, LTSpice, Mathematica, LaTeX, IDL, ROOT, Matlab, Mathematica
Operating Systems	Linux (Ubuntu, CentOS, Debian), Windows
Lab skills	RF design and measurement, cryogenics, ultra-high vacuum, laser optics, pulsed power, microcontrollers, signal processing, fast electronics, nanofabrication

Work Experience

Tibaray Inc., Fremont, CA, Senior R&D Engineer — 2024 - present

- Senior research and development engineer respondible for compact linac development for medical applications
- Design and fabrication of novel high power RF component including circulators, isolators, phase shifters, windows, and RF loads
- Multiphysics and CAD mechanical simulations
- RF component cold testing and quality management
- Linac vault operation

Particle Beam Physics Laboratory (PBPL), UCLA, Researcher — 2016 - 2024

- Commissioning of new CYBORG (CrYogenic Brightness-Optimized Radiofrequency Gun) beamline for photoemission and RF testing including development of S.O.P.
- Multiphysics simulation and RF cavity design and measurement
- Cryogenic hardware design for RF cavity accelerator
- Laser optics and vacuum engineering for high harmonic generation experiment
- Teaching and management of multiple teams of undergraduate research projects
- Plasmonic and beam dynamics simulations for surface studies involved in high harmonic generation
- Novel multipole magnet design
- Fabrication of nanoscales structures with anisotropic wet etches of silicon wafers

AEgIS Collaboration, CERN, Research Student 2015 - 2016

- Antiproton beam dynamics simulations
- Ion optics design and manufacturing: incl. einzel lenses, hemispheric analyzers, and Penning traps
- Commissioning of UHV beamline elements on Antiproton Decelerator

Education

· PhD in Physics **University of California, Los Angeles**
Thesis: *Improvements to Normal Conducting High Gradient Accelerator Performance at Cryogenic Temperatures*

· MS in Physics **University of California, Los Angeles**

· Coursework (4 courses, 12 credit hours) **US Particle Accelerator School**

· BA in Physics, *cum laude* with distinction **Boston University**
Senior Honors Thesis: *Design and Analysis of Permanent Magnet Penning Trap for AEGIS Collaborations Antiproton-beam Studies*

Selected Publications

Lawler, Gerard Emile et al. (2024). "Improving Cathode Testing with a High-Gradient Cryogenic Normal Conducting RF Photogun". In: *Instruments* 8.1. test 1. ISSN: 2410-390X. DOI: 10.3390/instruments8010014.

Rosenzweig, James B. et al. (2024). "A High-Flux Compact X-ray Free-Electron Laser for Next-Generation Chip Metrology Needs". In: *Instruments* 8.1. ISSN: 2410-390X. DOI: 10.3390/instruments8010019.

Lawler, Gerard et al. (2023). *Improving Interface Physics Understanding in High-Frequency Cryogenic Normal Conducting Cavities*. arXiv: 2310.11578 [physics.acc-ph].

Rosenzweig, J B et al. (Sept. 2020). "An ultra-compact x-ray free-electron laser". In: *New Journal of Physics* 22.9, p. 093067. DOI: 10.1088/1367-2630/abb16c.

Lawler, Gerard, Kunal Sanwalka, et al. (2019). "Electron Diagnostics for Extreme High Brightness Nano-Blade Field Emission Cathodes". In: *Instruments* 3.4. DOI: 10.3390/instruments3040057.

Rosenzweig, J.B. et al. (2018). "Ultra-high brightness electron beams from very-high field cryogenic radiofrequency photocathode sources". In: *Nucl Instrum and Meth A* 909. 3rd European Advanced Accelerator Concepts workshop (EAAC2017), pp. 224–228. DOI: 10.1016/j.nima.2018.01.061.